

One Dimensional Random Walk

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Abstract

The one-dimensional random walk is a fundamental model for exploring stochastic processes across various scientific domains. In this paper, we theoretically derive key statistical properties of the symmetric random walk, including the mean, standard deviation, and probability distribution of final positions. These derivations are complemented by numerical simulations, which confirm the analytical predictions and provide deeper insight into the convergence behavior as the number of steps increases. The results demonstrate the emergence of Gaussian probability distributions for large step counts, in accordance with the Central Limit Theorem. This study bridges theoretical analysis and computational validation, offering a comprehensive understanding of random walk dynamics with potential applications in physical and biological systems and even in stock market fluctuations.

1 Introduction

Random walks are a powerful tool for simulating and understanding various statistical phenomena in the natural world. In a random walk, a particle starts at a given position and moves in random directions over time. This concept has applications across numerous scientific and engineering disciplines, including ecology, psychology, computer science, physics, chemistry, biology, economics, and sociology. In physics, random walks serve as simplified models for processes like Brownian motion and diffusion, describing the random movement of molecules in liquids and gases. In mathematical ecology, they are used to model individual animal movements, support biodiffusion processes, and occasionally represent population dynamics. Numerical simulations allow us to compare random walk results with theoretical predictions, providing valuable insights into these systems. We would like to analyse these random walks and understand their statistical properties by answering questions like, what does the probability distribution of the position in a random walk look like? How does the standard deviation of the final position evolve with the number of steps in a random walk? Moreover, despite the inherent randomness of the process, can we identify any definiteness or patterns when the sample size (i.e., the number of steps) becomes very large? In this report, we aim to address these questions through theoretical analysis and numerical simulations.

2 Theoretical Understanding

From a mathematical perspective, a random walk is a stochastic process that involves a sequence of random steps on a mathematical space. Physically, this can be illustrated by a particle that moves randomly, taking a series of steps where the probability of moving in any direction is equally likely. The term "random walk" was first coined by Karl Pearson in 1905.[1] [2]

The simplest case involves a particle that starts at a position x_0 and moves randomly in either direction along the x -axis. In each step that the particle takes, the step length is fixed for the entire walk, and both directions (positive and negative) are equally likely. There can indeed be a biased random walk, such that the probability of taking either step is not equal, but here we limit ourselves mostly to a symmetric random walk. A symmetric random walk is defined by three variables; the initial position, the step length and the number of steps.

To theoretically analyse a one-dimensional random walk. We consider an arbitrary random walk with step length d , the walk will consist of a series of $+d$ and $-d$. Each step can be specified as X_i , where i runs from 1 to the number of steps (N). From the perspective of the probability theory, X_i is the **random variable**. and the set of all X_i is the **sample space**. The j th position x_j will be given by,

$$x_j = x_0 + \sum_{i=1}^{i=j} X_i \quad (1)$$

The expectation value of each step X_i will be,

$$\mathbb{E}[X_i] = (d \times 0.5) + (-d \times 0.5) = 0 \quad (2)$$

The 0.5 comes from the probability of each $+d$ and $-d$, since each one is equally likely. The Variance of each step can be found as,

$$\text{Var}(X_i) = \mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2 = \mathbb{E}[X_i^2] \quad (3)$$

X_i^2 is d^2 , because X_i is either $+d$ or $-d$. Thus $\mathbb{E}[X_i^2] = d^2$.

$$\text{Var}(X_i) = d^2 \quad (4)$$

Now we want to calculate the variance of x_N . Using the property that the variance of the sum of independent random variables is the sum of their variances, we get:

$$\text{Var}(x_N) = \text{Var}\left(x_0 + \sum_{i=1}^{i=N} X_i\right) = \text{Var}(x_0) + \sum_{i=1}^{i=N} \text{Var}(X_i) \quad (5)$$

The variance of a constant is zero.

$$\text{Var}(x_N) = d^2 \times N \quad (6)$$

The standard deviation σ will be given by,

$$\sigma(x_N) = d \times \sqrt{N} \quad (7)$$

Thus keeping the step distance fixed, the standard deviation of the random walks should increase with $\sqrt{N}$ and we shall observe it from numerical simulations. Also the expectation or the mean value of final position x_N can be found as,

$$\mathbb{E}[x_N] = \mathbb{E}\left[x_0 + \sum_{i=1}^{i=N} Z_i\right] \quad (8)$$

$$\mathbb{E}[x_N] = \mathbb{E}[x_0] + \sum_{i=1}^{i=N} \mathbb{E}[Z_i] \quad (9)$$

$$\mathbb{E}[x_N] = x_0 \quad (10)$$

Thus the mean value of the final position will be the initial position.

To analyse the probability distribution, consider that out of total N steps, N_1 steps are forward(+d) and N_2 steps are backwards(-d). Each random walk can be thought of as a unique collection of forward(+d) and backward(-d) steps. The total number of different random walks is 2^N and the total number of random walks with specific forward steps N_1 is given by,

$$g(N, N_1) = \frac{N!}{N_1!(N - N_1)!} \quad (11)$$

The probability of having exactly N_1 forward steps is given by,

$$P(N_1) = \frac{N!}{N_1!(N - N_1)!} \left(\frac{1}{2}\right)^N \quad (12)$$

The net displacement after N steps is $x = 2N_1 - N$, rearranging

$$N_1 = \frac{x + N}{2} \quad (13)$$

Substituting N_1 into $P(N_1)$ provides the probability of being at position x after N steps;

$$P(x) = \frac{N!}{\left(\frac{x+N}{2}\right)! \left(N - \frac{x+N}{2}\right)!} \left(\frac{1}{2}\right)^N \quad (14)$$

This probability distribution will tend towards a gaussian as N becomes very large due to the **Central Limit Theorem (CLT)**, which states that the sum (or average) of a large number of independent random variables, each with finite mean and variance, tends toward a normal (Gaussian) distribution.

Thus for large N , the probability distribution can be approximated by,

$$P(x, N) \approx \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right) \quad (15)$$

Where $\sigma = \sqrt{N}$. Thus, even though the process is random, for large number of steps, a pattern emerges. This pattern is the gaussian distribution function. The gaussian distribution in Eq(15) can be obtained from Eq(14) using stirling's approximation,

$$N! = (2\pi N)^{\frac{1}{2}} N^N \exp\left(-N + \frac{1}{12N} + O\left(\frac{1}{N^2}\right)\right) \quad (16)$$

Where $1/12N + O(1/N^2)$ becomes zero as N tends to ∞

3 Getting lost in Higher Dimensions

What we have analysed in the previous section, applies to a one dimensional case. But how does it relate to higher dimensions and how are higher dimensional random walks different from a one dimensional random walk? While the standard deviation and the probability distribution behaves similar to a one-dimensional walk, the distribution being three dimensional gaussian function. But, in some aspects the walks differ in different dimensions. A simple question one may ask is, what is the probability of a random walker that he/she returns to the starting position? George Poyla, a Hungarian-American mathematician proved in 1921 that if $p(d)$ is the probability that the random walker returns to his original position in a d dimensional space, then[3];

$$p(1) = p(2) = 1 \quad (17)$$

but for $d > 2$,

$$p(d) < 1 \quad (18)$$

Thus, in one and two dimensions the random walker will definitely cross his starting position again, but not in higher dimensions. For 3 dimensions, $p(3) = 0.34$, the probability drops down to 34% and keeps on decreasing as d increases[3]. The mathematician Shizuo Kakutani referred to this result with the following quote: "*A drunk man will find his way home, but a drunk bird may get lost forever*". Another interesting fact is that, in higher dimensions the random walk generates a discrete fractal, that is, a set which exhibits stochastic self-similarity on large scales. The probability $p(d)$ can be in general found from[4],

$$p(d) = 1 - \left(\frac{1}{\pi^d} \int_{[-\pi, \pi]^d} \frac{\prod_{i=1}^d d\theta_i}{1 - \frac{1}{d} \sum_{i=1}^d \cos(\theta_i)} \right) \quad (19)$$

4 Methods: Programming a random walk

4.1 A Single Random Walk

A single random walk can be constructed by generating a random sequence of $+d$ and $-d$ (*the step difference*).

$$Z = [+d \quad -d \quad -d \quad +d \quad -d \quad +d \quad +d \quad \dots \quad +d]$$

The number of these step differences will be the number of steps, N , that will be specified in our code. By incorporating the initial starting point x_0 into the generated sequence of steps, we obtain,

$$Z = [x_0 \quad +d \quad -d \quad -d \quad +d \quad -d \quad +d \quad +d \quad \dots \quad +d]$$

The random walk will be given by the cumulative sum of this series,

$$Z = [x_0 \quad Z_1 \quad Z_2 \quad Z_3 \quad Z_4 \quad Z_5 \quad Z_6 \quad Z_7 \quad \dots \quad Z_N]$$

That is our random walk with initial position x_0 , step difference d , and number of steps N .

A random walk with initial position 100, step difference 5 number of steps 200 has been plotted in Figure 1. The plot gives the position at each step of the walk.

4.2 Statistics of multiple random walks

To study the statistics of the final position of a random walk, we simulate multiple independent random walks, each consisting of the same number of steps. Each random walk is represented as a sequence of steps, and these sequences are stored in a matrix A , where the rows correspond to individual random walks, and the columns represent the step-by-step progression of each walk. Thus, the matrix A has dimensions equal to the number of walks (rows) number of steps (columns).

$$A = \begin{bmatrix} +d & -d & +d & \dots & +d \\ +d & +d & -d & \dots & -d \\ -d & -d & +d & \dots & -d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -d & +d & +d & \dots & +d \end{bmatrix}$$

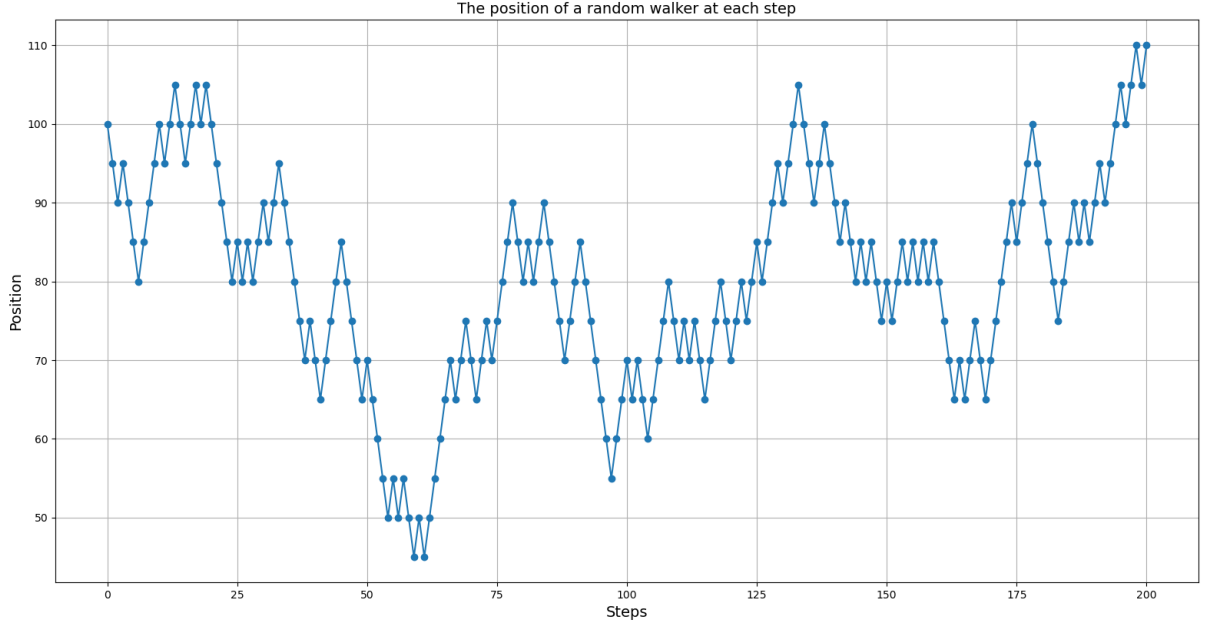


Figure 1: Plot of the position at each step in a random walk

Adding the starting point x_0 in the first column of the matrix and evaluating the cumulative sum for each row, we obtain our matrix A consisting of multiple random walks all with same number of steps N .

$$A = \begin{bmatrix} x_0 & Z_{11} & Z_{12} & Z_{13} & \cdots & Z_{1N} \\ x_0 & Z_{21} & Z_{22} & Z_{23} & \cdots & Z_{2N} \\ x_0 & Z_{31} & Z_{32} & Z_{33} & \cdots & Z_{3N} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ x_0 & Z_{M1} & Z_{M2} & Z_{M3} & \cdots & Z_{MN} \end{bmatrix}$$

Here Z_{ij} will be the position at j th step in i th random walk.

Here:

- Each row corresponds to the position history of one walk.
- The cumulative sums across columns in each row represent the walker's position at each step.

This structure allows for efficient statistical analysis, such as calculating the mean and variance of the positions across multiple random walks, particularly focusing on the distribution of the final positions (last column of A). We then compute the average or standard deviation or any other statistical quantity of interest using the final positions of each walk, which corresponds to the last column in the matrix A .

To observe the behaviour of standard deviation or any other statistical variable of the final positions with respect to the number of steps, we vary the number of steps. Thus, creating multiple matrices A , each with a different number of steps, starting from zero and going upto a fixed value, specified in the code.

Collecting the standard deviations from all the matrices we can analyse its variations with the number of steps.

5 Results/Discussion: Observations from numerical simulations

There are various statistical quantities that can be derived from simulations of random walks to better understand their behavior. One of the most fundamental questions is: What is the probability of a particular value of x being the final position of a random walk? From theoretical considerations and Equation (15), we know that for a large number of steps (N), the distribution of the final positions should approximate a Gaussian distribution due to the Central Limit Theorem. To verify this numerically, we simulate multiple independent random walks. As described earlier, we organize the simulations in a single matrix A , where:

- Each row represents the trajectory of an individual random walk.
- Each column represents the positions across different steps.

In this example: The starting position for all walks is set to 100 ($x_0 = 100$). The number of random walks is set to 1000, providing a sufficiently large sample to accurately estimate the statistical distribution. By plotting this empirical probability distribution, we can compare it with the theoretical Gaussian prediction. For sufficiently large N , the match between the simulated data and the Gaussian curve should confirm the theoretical result.

This simulation not only validates the Gaussian nature of the distribution but also provides insights into the variability of outcomes in random processes. This behaviour of probability distributions is very general and not just limited to random walks, but is a general feature of any probability distributions.

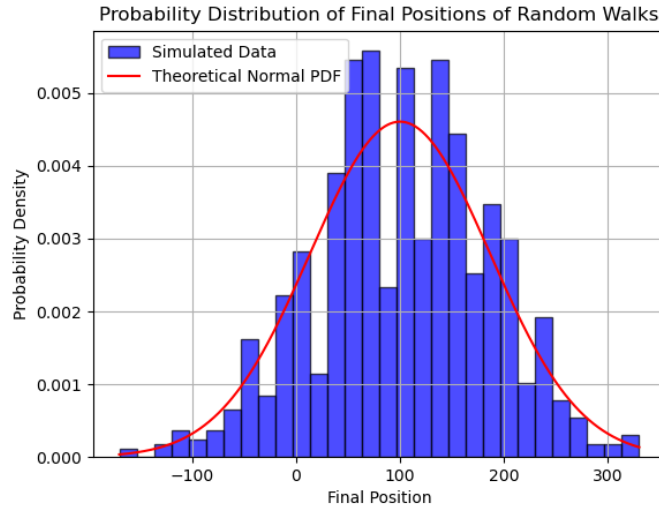


Figure 2: Probability distribution, Number of steps=300

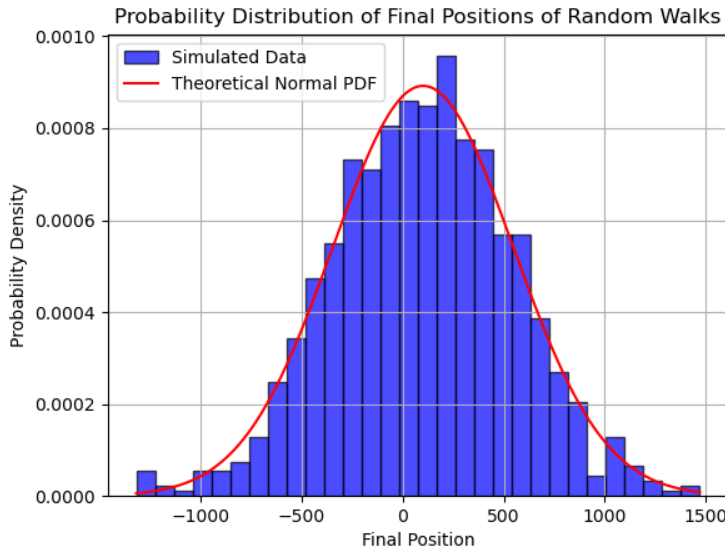


Figure 3: Probability distribution, Number of steps=7000

The difference between the two plots can be observed (Fig 2 and 3). For higher number of steps, the distribution is better approximated by a gaussian.

We can also analyse the variation of standard deviation with the number of steps. Figure 4, shows the evolution of the standard deviation with the number of steps, and it matches well with the theoretical derivations from section 2. We can also observe that the expectation value of the final position fluctuates around the initial position (zero).

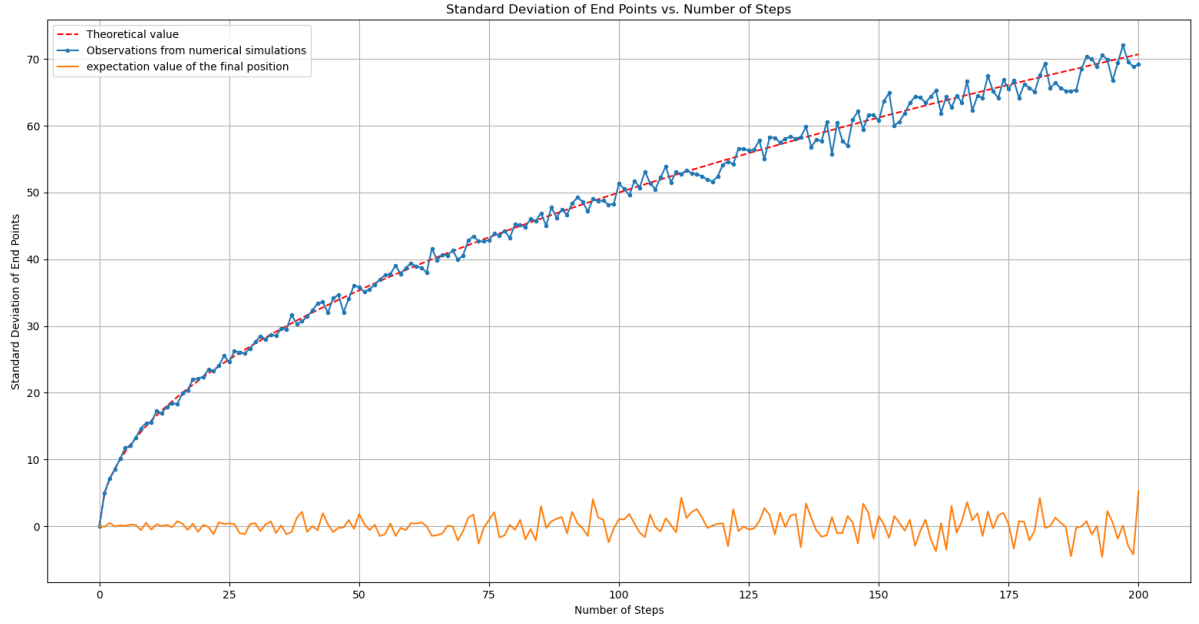


Figure 4: Evolution of the standard deviation with the number of steps

We can observe from this plot that the fluctuations in standard deviation increase with the number of steps. We can quantify these fluctuations by running the simulation multiple times, then finding the standard deviation of the standard deviations and plotting with error bars.

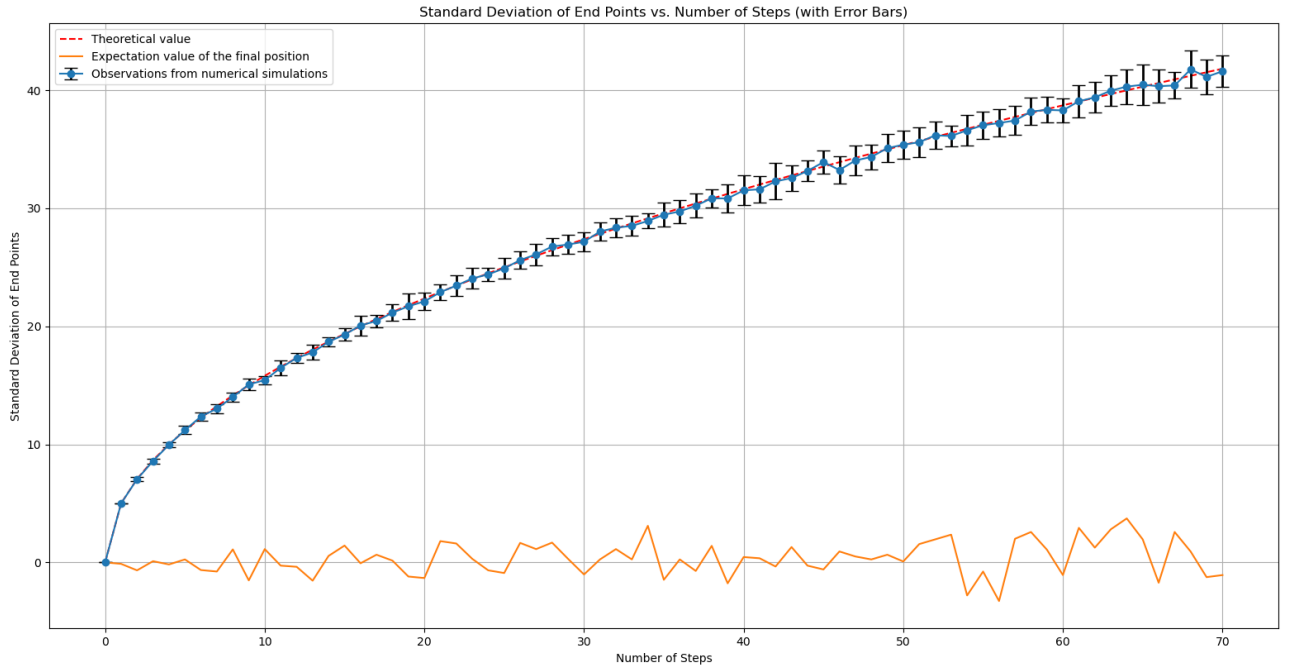


Figure 5: Evolution of the standard deviation with the number of steps with the error bars

6 Conclusion

This paper's analysis highlighted the fundamental statistics of random walks. Theoretical predictions showed that the probability distribution of a random walk approaches a Gaussian as the number of steps increases. Additionally, the relationship between the standard deviation and the number of steps was derived. Both theoretical results were validated through numerical simulations. This approach provides a clear understanding of random walk properties through computational methods, without the need for direct experimentation. Such insights are invaluable for modeling various physical phenomena involving random particle motion, such as gas diffusion, particle transport in biological systems, and stock market fluctuations.

7 References

References

- [1] Eric W. Weisstein. Random walk–1-dimensional. *From MathWorld–A Wolfram Web Resource*.
- [2] Feller William. *An Introduction To Probability Theory and Its Applications*.
- [3] Eric W. Weisstein. Pólya's random walk constants. *From MathWorld–A Wolfram Web Resource*.
- [4] Kenneth Lange. Polya's random walk theorem revisited. *The American Mathematical Monthly*, 2015.